Reproducing the results of the paper

```
clear all; close all; clc;
```

Load the dataset

```
x1 = [0.1,0.3,0.1,0.6,0.4];
y1 = [0.1,0.4,0.5,0.9,0.2];

x2 = [0.6,0.5,0.9,0.4,0.7];
y2 = [0.3,0.6,0.2,0.4,0.6];

scatter(x1, y1, 'blue')
hold on;
scatter(x2, y2, 'red')

x = [x1, x2; y1, y2];
y = [ones(1,5) zeros(1,5); zeros(1,5) ones(1,5)];
```

Define the hyperparameters of the network:

- `shape` contains the number of neuron per layer of the network
- `sigma` and `sigmaprime` are the activation function of the network, here we use sigmoid
- `niter` is the maximum number of iterations
- `eta` is the learning rate

```
sigma = @(t) 1./(1+exp(-t));
sigmaprime = @(t) sigma(t).*(1-sigma(t));
shape = [2, 3, 3, 2];
niter = 3e5;
eta = 0.05;
```

Training

```
[costHistory, W, b] = GradientDescentClassification( ...
        x, y, niter, sigma, sigmaprime, eta, shape);
```

```
10.0%
20.0%
30.0%
40.0%
50.0%
60.0%
70.0%
80.0%
90.0%
100.0%
```

```
save NNparams.mat W b
```

Plot the evolution of the cost function against the iterations.

```
figure
plot(linspace(0,niter,niter)', costHistory, '-')
```

```
fprintf('Cost Function: %f\n', costHistory(end));
```

```
Cost Function: 0.000348
```

Prediction

Generate some other data in the domain to test the performance of the network.

```
load NNparams.mat
% generate new random data in the square [0,1]^2
n = 300;
testDatax = rand(n, 2);
testDatay = PredictClasses(W, b, sigma, testDatax');

x = x';
y = int32([zeros(5, 1); ones(5, 1)]');
figure
scatter(x(:,1), x(:,2), [], y, 'o');
hold on;
scatter(testDatax(:,1), testDatax(:,2), [], testDatay, 'filled');
```

Testing our implementation in a more involved case

Here we use a synthetic dataset that has been generated using the python script reported in the repository. The dataset contains points in R^2 clustered in 4 different classes.

Pre-processing

```
clear all; close all; clc;
```

Load the dataset and set its parameters

```
load X_new.mat; load y_new.mat; y = Y;
% normalize dataset
% m = mean(X);
% s = max(X);
% for i = 1:2
%     X(:,i) = (X(:,i) + m(i))/s(i);   % normalize every dimension
% end

n = size(X, 1);
d = size(X, 2);
n_classes = 4;
```

Visualize the dataset

```
figure
scatter(X(:, 1), X(:, 2), [], y);

% transform y to categorical (inefficient but needed)
y_cat = zeros(4, n);
for k = 1:n
    y_cat(y(k)+1,k) = 1;
end
```

Split the dataset into training and test

```
split_ratio = 0.3;
```

```
k = floor(split_ratio*n);

X_train = X(1:k,:);
X_test = X(k+1:end,:);
y_train = y(1:k);
y_test = y(k+1:end);
y_train_cat = y_cat(:,1:k);
y_test_cat = y_cat(:,k+1:end);
```

Define some hyperparameters of the network

```
sigma = @(t) 1./(1+exp(-t));
sigmaprime = @(t) sigma(t).*(1-sigma(t));

shape = [d, 3, 3, n_classes];

niter = 3e5;
eta = 0.05;
```

Training

```
[costHistory, W, b] = GradientDescentClassification( ...
        X_train', y_train_cat, niter, sigma, sigmaprime, eta, shape);
```

```
10.0%
20.0%
30.0%
40.0%
50.0%
60.0%
70.0%
80.0%
90.0%
100.0%
```

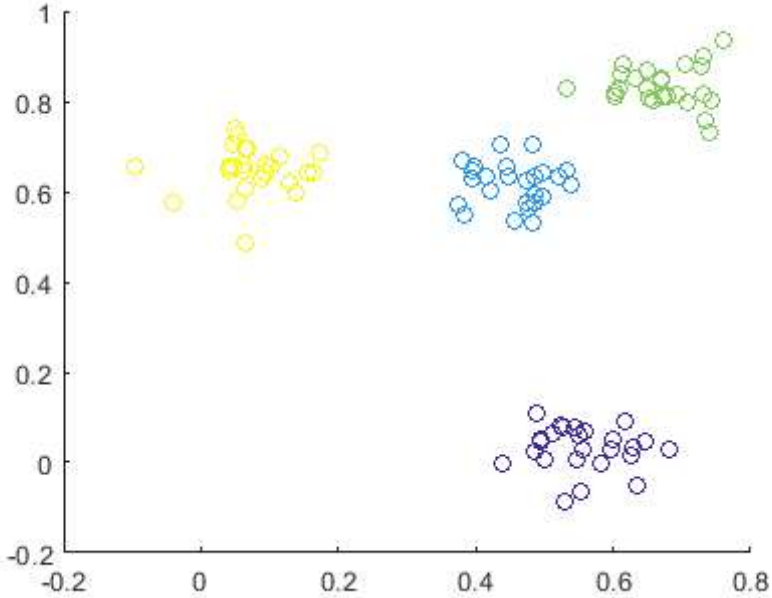
```
save NNparams.mat W b
```

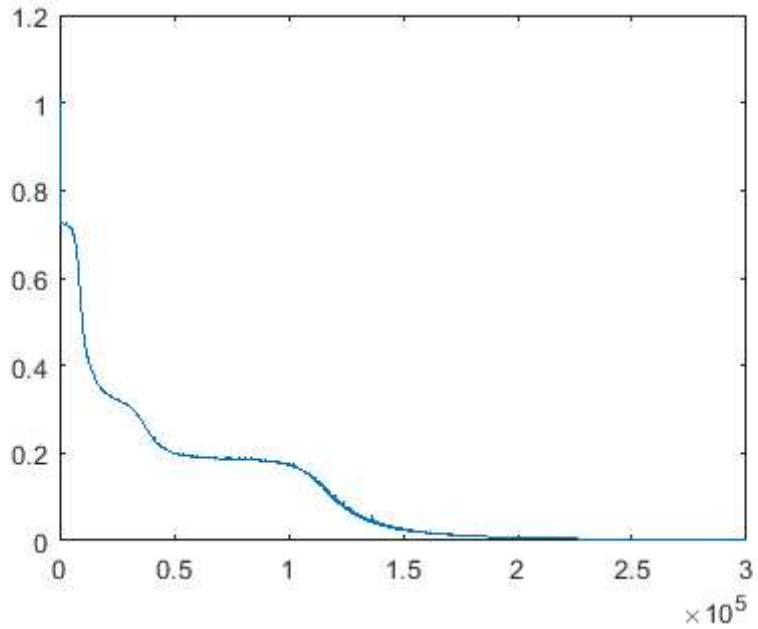
Plot `costHistory` as done before (and save `png` file)

```
figure
plot(linspace(0,niter,niter)', costHistory, '-')
fprintf('Cost Function: %f\n', costHistory(end));
```

```
Cost Function: 0.002174
```

```
filename = 'training_cost_function.png';
saveas(gcf, filename, 'png');
```





Test

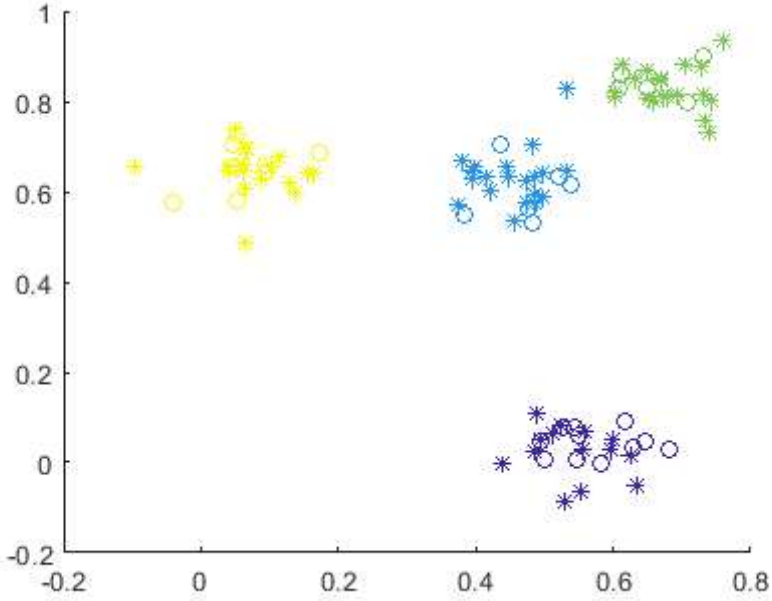
```
load NNparams.mat W b
```

Predict the class of the points in the test set using the network

```
y_pred = PredictClasses(W, b, sigma, X_test');
```

Plot the prediction

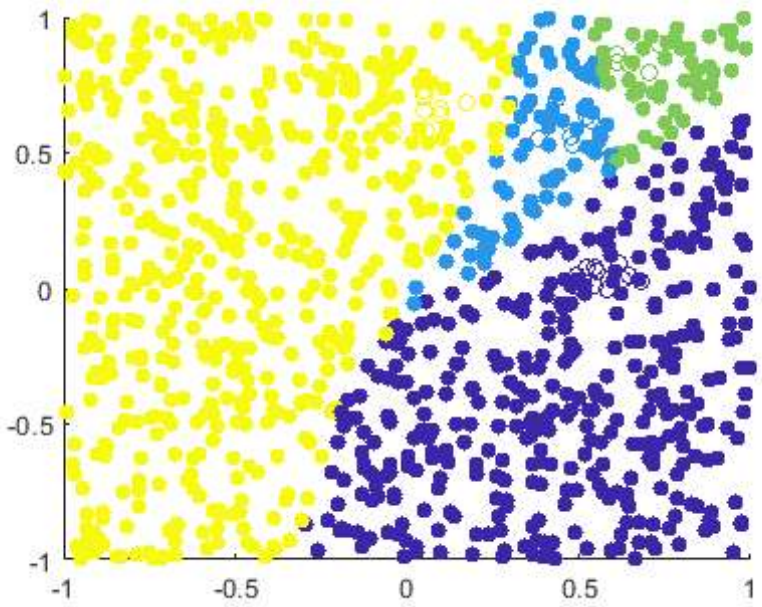
```
figure
scatter(X_train(:, 1), X_train(:, 2), [], y_train, 'o');
hold on;
scatter(X_test(:, 1), X_test(:, 2), [], y_pred, '*');
saveas(gcf, '4_classes.png', 'png');
```




```
% confusion matrix
figure
y_test = int32(y_test);
confusionchart(y_test,y_pred)
saveas(gcf, 'ConfusionMatrix', 'png');
```

Boundary exploration

We'd like to understand the criterion the network has learned to categorize the points. Therefore we generate a set of points that is suppose to fill the whole domain, and see how the network classifies the latter.

```
load NNparams.mat W b
% generate new random data in the square [0,1]^2
n = 1000;
testDatax = -1 + 2*rand(n, 2);
testDatay = PredictClasses(W, b, sigma, testDatax');
```

```
figure
scatter(X_train(:,1), X_train(:,2), [], y_train, 'o');
hold on;
scatter(testDatax(:,1), testDatax(:,2), [], testDatay, 'filled');
saveas(gcf, 'SpaceExploration', 'png');
```